

Ride-Hailing Demand & Smart Charging for an Electric Fleet

AAA Team Assignment, SS 2026

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1 Problem Description

Business context and goal. A renowned German car manufacturer is establishing a ride-hailing platform operated with a fully electrified vehicle fleet, targeting the more receptive US market first. While the client masters vehicle manufacturing, it lacks the tactical and strategic know-how to operate an electric ride-hailing fleet. As the data science team of Bane & Wayne Partners, our goal is to give the client a data-driven understanding of ride-hailing demand for a representative US city in both spatial and temporal resolution, and to translate that understanding into fleet decisions on three levels. The strategic level concerns which districts to enter first. The tactical level concerns how many vehicles to deploy and at what times and locations. The operational level concerns how to charge and reposition the vehicles. Because the fleet is fully electric, charging is not an afterthought but a central cost and sustainability lever: shifting charging into the right hours lowers operating cost and eases grid load, which fits the course’s focus on socially responsible, sustainability-oriented analytics. Since no proprietary ride-hailing data exists yet, we use the City of Chicago taxi trip records (2024 onwards) as a proxy for ride-hailing demand. Taxis are a reasonable stand-in, although they differ from app-based hailing in booking behaviour and market coverage, a limitation we keep in view when drawing conclusions.

Data-mining goal. We translate this business goal into a sequence of analytics problems that together form the project. (1) *Data preparation*: turn the large, privacy-restricted raw trip records into clean, spatially and temporally aggregated demand panels, discretising the city with a hexagonal H3 grid and enriching it with external weather data. (2) *Descriptive spatial analytics*: characterise how demand and trip patterns vary across space and time at different resolutions, and locate demand hotspots using Gaussian Mixture Models and kernel density estimation. (3) *Predictive analytics*: frame demand as a supervised regression of the target `demand_count` per spatial unit and time bucket, conditioned on calendar, weather and location. This answers questions such as “what demand can we expect on a rainy Sunday from 10 to 11am at the outskirts of the city”, rather than forecasting the next hour from the current one, and it uses Support Vector Machines and feed-forward neural networks. (4) *Prescriptive analytics*: design a smart-charging agent via reinforcement learning that minimises recharging cost while avoiding the risk of running out of energy during the working day. (5) *Synthesis*: consolidate the findings into actionable recommendations for the client, including whether the operator should rely on private or public charging infrastructure. A guiding question throughout is cost-effectiveness. We benchmark each model against simpler baselines and ask whether the added

complexity, such as deep learning over a kernel SVM, is worth it for the client. Success is measured with standard regression metrics (MAE, RMSE, R^2) relative to a naive benchmark for the predictive models, and by recharging cost versus an energy-shortfall penalty for the charging agent.